



Aleef: An AI-Based Platform for Pet Adoption

Computer Science

Buthaynah Alsharef
Aghadi Alrebdi

422205680
422205807

Raseel Alnuwaysir
Wateen Alsaif

412205570
421202072

Supervisor: Dr. Muna Almushyti

Introduction

Finding the right pet, caring for it, or recovering a lost pet can be difficult using traditional methods.

Aleef is an AI-powered platform that combines pet adoption, lost pet reporting, veterinary services, and intelligent recommendations in one application.

It helps users make better adoption decisions while improving animal welfare.

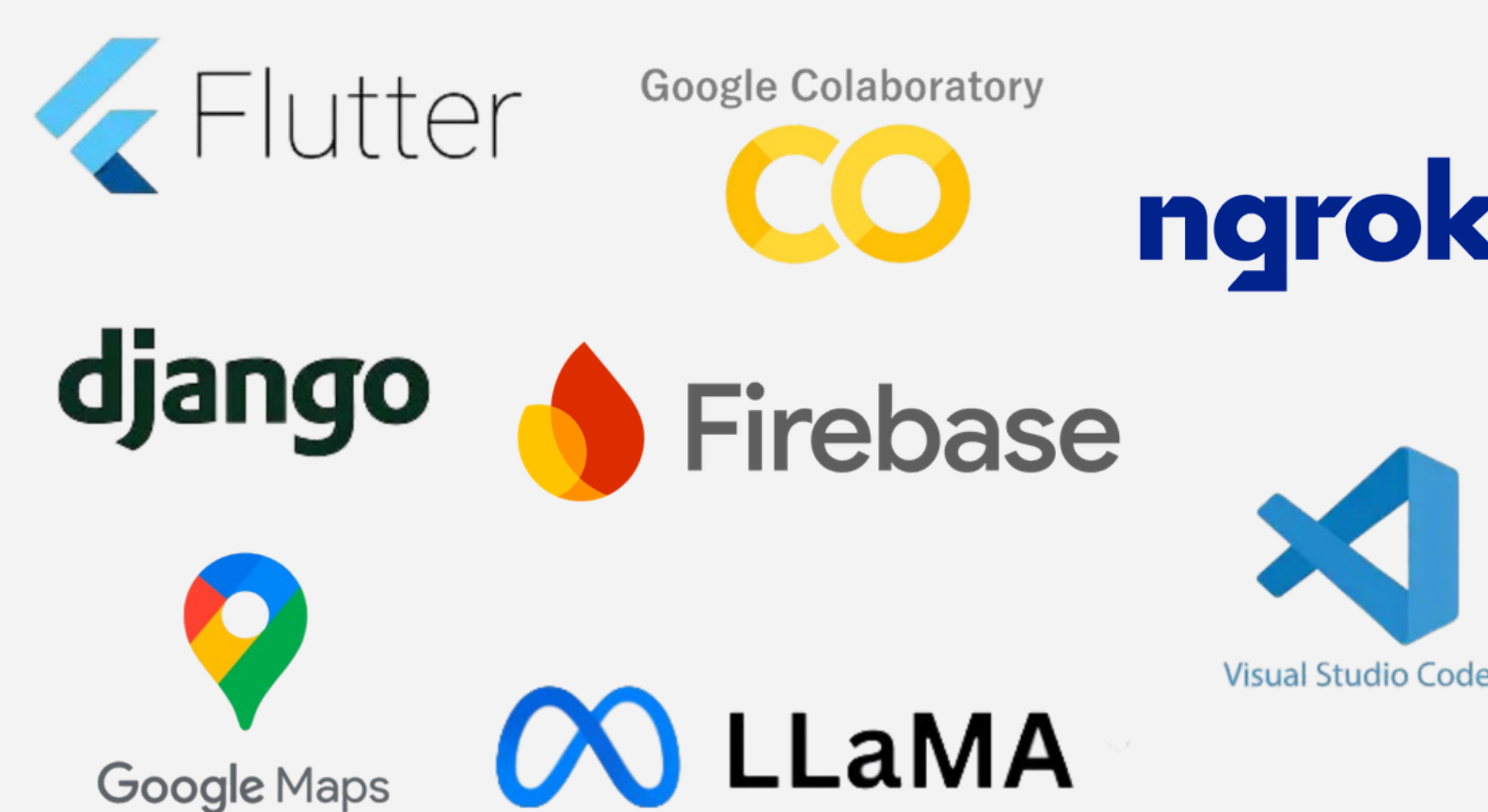
Problem

- No digital pet ID system for tracking and identification.
- Pet adoption lacks a unified system, causing inefficiency and lost pets.
- Social media is unreliable for adoption and lost pet cases.
- The pet community is fragmented with no centralized resources or instant communication.

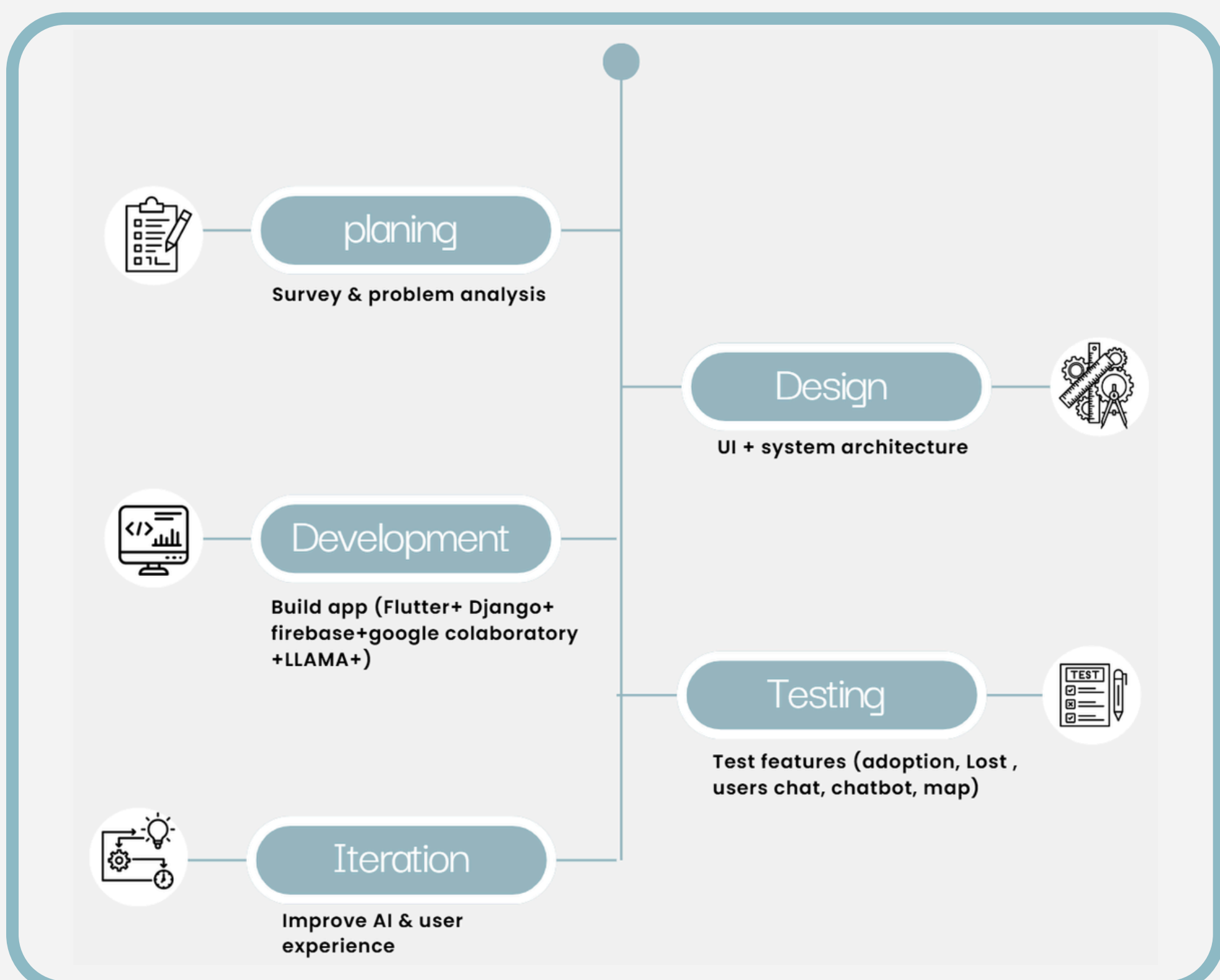
Solution

- Digital Pet ID System
- Unified Pet Adoption Platform
- Smart Hybrid Recommendation algorithms system
- Report Lost & Found Pets
- Nearby Vet Clinics
- Instant Chat with Owners & Vets
- AI Chatbot for Smarter Care

Implementation tools

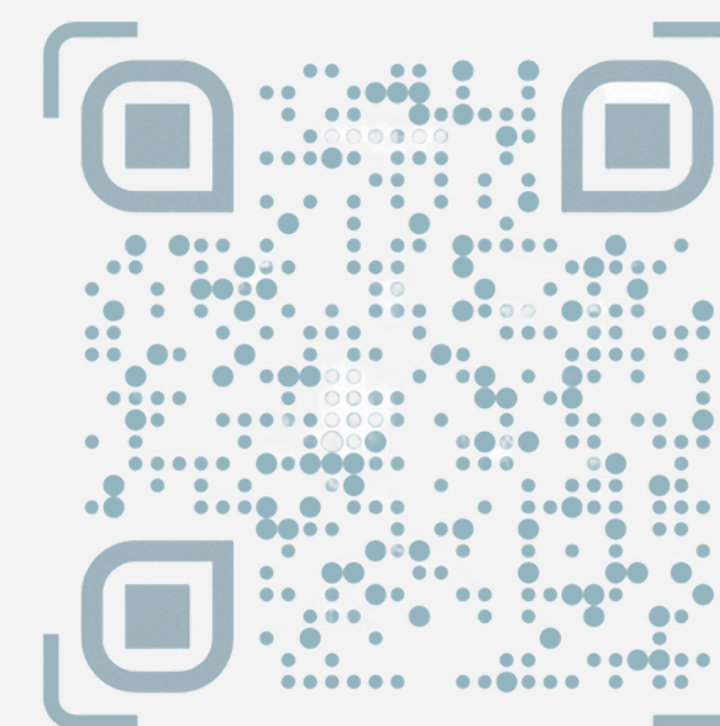


Methodolgy



Testing

- Recommendation engine: KNN, content-based filtering, constraint rules
- Chatbot: LLaMA (LLM) for user interaction
- Firestore data processing: One-Hot Encoding & normalization
- Weighted scoring model for balanced recommendations
- Focus on similarity, age, health & location factors
- Personalization, chatbot responses & system efficiency



Demo